

Eunseong Choi, Ph.D.

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Summary

I'm a Research Scientist at Liner. My research lies in **Information Retrieval** and **Natural Language Processing**, with an emphasis on building **robust and efficient Retrieval-Augmented Generation** frameworks. My work has received 300+ citations. Representative research includes:

- **Retrieval-Augmented Generation:**
 - Context–memory conflicts [C1]
 - Prompt compression for efficient LLMs [C4]
- **Information Retrieval:**
 - Passage re-ranking [C2]
 - Learned sparse representation [C8]
 - Generative recommendation [C3]
- **Natural Language Processing:**
 - Metaphor detection [C10]
 - Query reduction [C7]
- **Machine Learning:**
 - Knowledge tracing [C6]
 - Extreme multi-label classification [C9]

Experience

- 2025.12 – Current  **ML Research Scientist**, Core Tech Division, Liner
- Trained and deployed a neural re-ranker, improving retrieval performance by 12.5% over the previous production model
 - Built a daily incremental pipeline to improve search index freshness
- 2025.3 – 2025.4  **Research Intern**, Search & Ranking Modeling, NAVER
- Worked on continual fine-tuning for recency-aware dense retrieval
- 2021.7 – 2021.8  **Research Intern**, Search CIC, NAVER
- Developed a sparse retrieval method for efficient first-stage retrieval

Education

- 2020.3 – 2026.2  **M.S./Ph.D., Artificial Intelligence**, Sungkyunkwan University
Advisor: Prof. Jongwuk Lee
Thesis: *Improving Evidentiality and Compression for Retrieval-Augmented Generation*
- 2012.3 – 2020.2  **B.S., Architecture**, Sungkyunkwan University
 **B.S., Samsung Convergence Software Course**, Sungkyunkwan University

Publications

Eunseong Choi, June Park, Hyeri Lee, Jongwuk Lee

- Mitigated context–memory conflicts in retrieval-augmented generation using adversarial soft prompting

[C2] Multi-view-guided Passage Reranking with Large Language Models *EMNLP 2025*
Jeongwoo Na, Jun Kwon, **Eunseong Choi**, Jongwuk Lee

- Proposed efficient and robust LLM-based passage reranker using multi-view embeddings

[C3] GRAM: Generative Recommendation via Semantic-aware Multi-granular Late Fusion *ACL 2025*
Sunkyoung Lee, Minjin Choi, **Eunseong Choi**, Hye-young Kim, Jongwuk Lee

- Efficiently leveraging multi-granular semantic information for sequential recommendation

[C4] From Reading to Compressing: Exploring the Multi-document Reader *EMNLP 2024 Findings*
for Prompt Compression
Eunseong Choi, Sunkyoung Lee, Minjin Choi, June Park, Jongwuk Lee

- Prompt compression using cross-attention, reducing prompt length by 80% while preserving global context

[C5] Multi-granularity Guided Fusion-in-Decoder *NAACL 2024 Findings*
Eunseong Choi, Hyeri Lee, Jongwuk Lee

- Improved open-domain QA by leveraging evidence at multiple levels of granularity

[C6] Forgetting-aware Linear Bias for Attentive Knowledge Tracing *CIKM 2023 (short paper)*
Yoonjin Im*, **Eunseong Choi***, Heejin Kook, Jongwuk Lee

- Modeled forgetting behavior with a linear bias in attention-based models

[C7] ConQueR: Contextualized Query Reduction using Search Logs *SIGIR 2023 (short paper)*
Hye-young Kim*, Minjin Choi*, Sunkyoung Lee, **Eunseong Choi**, Young-In Song, Jongwuk Lee

- Query reduction by combining core term extraction and sub-query selection from real-world search logs

[C8] SpaDE: Improving Sparse Representations using a Dual Document Encoder for *CIKM 2022*
First-stage Retrieval
Eunseong Choi*, Sunkyoung Lee*, Minjin Choi, Hyeseon Ko, Young-In Song, Jongwuk Lee

- Improved sparse retrieval by jointly learning term weighting and semantic expansion with a dual encoder

[C9] Long-tail Mixup for Extreme Multi-label Classification *CIKM 2022 (short paper)*
Sangwoo Han, **Eunseong Choi**, Chan Lim, Hyunjung Shim, Jongwuk Lee

- Addressed label sparsity in extreme classification with mixup-based augmentation

[C10] MelBERT: Metaphor Detection via Contextualized Late Interaction using *NAACL 2021*
Metaphorical Identification Theories
Minjin Choi, Sunkyoung Lee, **Eunseong Choi**, Heesoo Park, Junhyuk Lee, Dongwon Lee, Jongwuk Lee

- Metaphor detection using contextualized word representations and linguistic theories

Academic Service

Reviewer ■ ARR 2025 May (EMNLP 2025)

External Reviewer ■ EMNLP 2022; ACL 2023; SIGIR 2023–2024; KDD 2022, 2026; WSDM 2022; ARR 2025 Feb (ACL 2025)

Honors & Awards

2024 🏆 **1st Prize, Best Graduate Research Paper Award**, Sungkyunkwan University.
Presented research on prompt compression for efficient LLM

2023 🏆 **2nd Place (Minister's award), AI Grand Challenge for Policy Support AI**, IITP.
Developed a multi-hop retriever and re-ranker for generating policy-supportive documents

Honors & Awards (continued)

- 2022 🏆 **1st Place (Minister's award), AI Grand Challenge for Math Word Problem Solving, IITP.**
Developed a framework for math word problem solving using domain-specific operations
- 2021 🏆 **2nd Prize, Best Graduate Research Paper Award, Sungkyunkwan University.**
Presented research on efficient sparse retrieval using term expansion and re-weighting

Scholarships

- 2020 – 2022 **Graduate School Scholarship**, Department of AI, Sungkyunkwan University
- 2019 **Academic Excellence Scholarship**, Sungkyunkwan University
- 2017 – 2019 **National Scholarship for Science and Engineering Students**, Korea Student Aid Foundation
- 2015 – 2016 **Academic Excellence Scholarship**, Sungkyunkwan University.

Teaching Assistant

Regular Courses

- Spring 2023 **Introduction to Recommender Systems**
- Fall 2022 **Fundamentals of Machine Learning**
- Fall 2021 **Deep Neural Networks**
- Spring 2020 – 2022 **Introduction to Database**

Extracurricular Courses

- Machine Learning **LG Electronics** May 2020, Oct 2020, May 2021, Oct 2021, Oct 2022, Oct 2024
Samsung SDS Jun 2020
SK Innovation Sep 2020
Machine learning basics, including regression and neural networks
- NLP Project **Samsung SDS** Sep 2022
Covered topic classification and machine reading comprehension

Industry–Academia Projects

- 2025.09 – 2026.02 **NL2SQL-based Intelligent Question Answering System Development**
WeZON *Project Manager* - Developed an NL2SQL system enabling non-experts to query databases in natural language
- 2021.07 – 2022.06 **Document Retrieval via Pre-trained Language Models and Inverted Index**
NAVER *Team Leader* - Designed a dual-document encoder for effective sparse representations
- 2020.05 – 2021.04 **Topic-aware Query–Document Matching with Deep Neural Networks**
NAVER Leveraged VAE-based topic modeling to represent query–document semantics

References

Available upon request